

RESEARCH ARTICLE

Phototunable Nonreciprocal Cilia-Like Self-Oscillator

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ABSTRACT

Natural cilia generate non-reciprocal beating through coordinated deformation and recovery, enabling sensing, propulsion, transport, and feeding. Synthetic cilia-like actuators have been widely explored, yet combining self-sustained oscillation, non-reciprocal motion, multimode tunability, and adaptive behavior in fluidic environments remains a daunting challenge. Here we report a phototunable nonreciprocal cilia-like self-oscillator (PNCOS) that address this challenge. PNCOSs are based on light-driven high-degree-of-freedom deformation of self-winding liquid crystal elastomer fibers, which can generate multiple non-reciprocal oscillation modes through coupled 3D deformation and dynamic stiffness modulation. PNCOSs enable real-time tuning of oscillatory motion through multiple optical inputs. Interestingly, two adjacent PNCOSs can communicate with each other, and give rise to synchronized, collective motion through compliant coupling. In addition, their self-sustained non-reciprocal oscillation enables active manipulation of small objects in viscous aqueous environments. This work establishes an effective strategy for constructing adaptive and functional artificial ciliary systems for soft robotic and future microfluidic applications.

1 | Introduction

Cilia are ubiquitous motile organelles that drive propulsion and fluid transport in diverse biological systems [1, 2]. Because they typically operate at low Reynolds number, where viscous forces dominate and inertia is negligible, reciprocal motion cannot generate net flow. Effective propulsion therefore requires non-reciprocal beating cycles that break time-reversal symmetry [3]. For natural cilia, this motion asymmetry is commonly manifested by distinct effective and recovery strokes that trace different trajectories and define a finite swept area, thereby producing directional fluid transport [4]. Reproducing such non-reciprocal beating is thus a central requirement for cilia-like actuators intended for fluid manipulation in viscous environments.

Inspired by the versatile functions of natural cilia, artificial cilia-like actuators have been widely developed for applications in microfluidics [5–8], microsensing [9–11], and microrobotics

[12–14]. Although these systems can be actuated by diverse external stimuli, including electric [15], magnetic [16, 17], pneumatic [18], and optical fields [19], reproducing the non-reciprocal beating of natural cilia remains challenging. Current strategies mainly rely on elastic instability [20] or multisegment actuation [21] to generate spatial asymmetry. However, both approaches typically require sophisticated external control, increasing system complexity and limiting robustness, and therefore remain difficult to integrate with autonomous, self-sustained operation.

This challenge becomes even greater in fluidic environments, where viscous damping suppresses motion and makes sustained oscillation difficult to maintain. Moreover, even when self-oscillation can be achieved, reciprocal motion is generally inefficient for generating directional transport in viscous fluidic environments, particularly under low-Reynolds-number conditions [22–25]. Recent studies have shown that underwater non-reciprocal self-oscillation can be realized by orthogonal

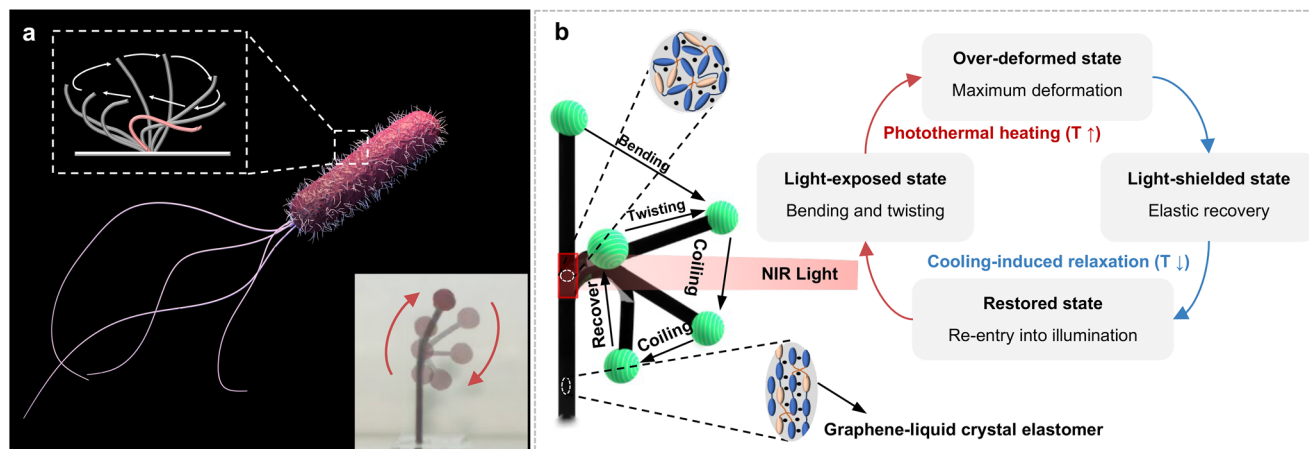


FIGURE 1 | Concept of a light-powered nonreciprocal self-oscillation system. a) Schematic representation of natural ciliary movement. The inset at the upper left shows the 3D beating trajectory of a natural cilium, and the inset at the lower right shows the 3D motion trajectory of the artificial cilium under NIR irradiation. b) Schematic illustration of an cilia-like self-oscillator, showing light-driven out-of-equilibrium motion. The inset illustrates the photothermal-mechanical feedback cycle that sustains self-oscillation.

optical excitation [26], enabling cilia-like motion, fluid pumping, and micro-object transport. However, creating biomimetic cilia-like actuators, capable of integrating non-reciprocal beating with autonomous, self-sustained oscillation, multimode tunability, and adaptive behavior in fluidic environments, yet has been pursued as a desired goal in the community of smart materials and soft robotics.

Here, we report phototunable nonreciprocal cilia-like self-oscillators (PNCsOs) based on a self-winding liquid crystal elastomer (LCE) fiber actuator. Functioning as biomimetic cilia-like actuator in fluidic environments, this cilia-like system integrates two essential features. First, light-induced changes in molecular order dynamically modulate the mechanical properties of the actuator, leading to distinct responses during the deformation and recovery phases. Second, an asymmetric stress distribution across the fiber cross-section drives a reversible straight-to-helical transformation, enabling active deviation from the incident light and the establishment of a self-feedback loop for sustained oscillation. Together with a reduced nematic-to-isotropic transition temperature that accelerates actuation in fluidic environments, these features enable autonomous self-oscillation and non-reciprocal beating within a single actuation system.

2 | Results and Discussion

2.1 | Design and Fabrication of a Bioinspired Oscillatory Cilium

The PNCsO introduced here is designed to autonomously convert constant near-infrared (NIR) illumination into asymmetric oscillatory motion in aqueous environments. The basic architecture consists of three components: (1) a self-winding LCE fiber that serves as active deformable body, (2) a buoyant foam bead attached to the tip of the LCE fiber to provide an upward force and regulate the bent configuration in water, and (3) an NIR light source that supplies photothermal input for actuation and control (Figure 1b). This concise fiber-bead configuration

establishes the geometric and mechanical basis for autonomous cilia-like oscillation, in which light-induced deformation of the LCE fiber, buoyancy-assisted recovery, and position-dependent illumination are coupled to generate non-reciprocal beating.

Inspired by the efficient beating behavior of natural cilia, which achieve non-reciprocal motion through coordinated bending, elastic recovery, and phase-dependent mechanical responses (Figure 1a), this design was guided by two key features of natural ciliary motion: elastic energy storage and release during bending-recovery cycles, and distinct mechanical responses during the deformation and recovery phases, which underlie non-reciprocal beating [27]. To enable rapid actuation in aqueous environments, liquid crystal diacrylate (C6BAPE) was incorporated into the thiol-Michael addition network (Figure S1), which lowered the nematic-to-isotropic transition temperature while preserving a large deformation strain (Figures S2 and S3) [28, 29]. Graphene was further introduced to enhance the mechanical robustness and photothermal responsiveness of the LCE matrix, thereby promoting rapid light-driven deformation in aqueous environments [30, 31]. These features established the material basis for a self-winding cilia-like actuator capable of elastic recovery, sustained oscillation, and non-reciprocal beating.

In addition to rapid actuation in aqueous environments, PNCsOs require an internal feedback mechanism to sustain oscillation under constant illumination. Our previous studies showed that self-winding LCE fibers undergo reversible linear-to-helical transformations under near-infrared (NIR) irradiation, accompanied by pronounced deformation and shape evolution (Figure S4) [32, 33]. Under partial irradiation, the resulting asymmetric photothermal deformation drives the fiber away from the illuminated region. Once removed from the light field, the fiber cools and recovers, returning it toward the irradiation zone. Repetition of this light-driven deformation-recovery cycle establishes a self-sustained photothermal-mechanical feedback loop, making self-winding LCE fibers a suitable platform for constructing biomimetic cilia-like actuators with autonomous oscillation (Figure 1b).

Non-reciprocal beating further requires distinct deformation and recovery paths. In natural cilia, this asymmetry is associated with phase-dependent mechanical responses, in which the recovery stroke is softer and more strongly affected by hydrodynamic resistance [34]. In our design, non-reciprocity was introduced by integrating a buoyant element at the apex of the cilia-like actuator. Upon irradiation, the initially linear cilium transformed into a helical configuration through coupled bending, twisting, and coiling deformations, generating an arc-shaped actuation path. Meanwhile, disruption of the ordered liquid-crystal mesogens reduced the stiffness of the LCE under illumination. The combination of light-induced softening and buoyancy shortened the recovery path relative to the actuation path, thereby producing a non-reciprocal motion trajectory.

To encode the asymmetric strain and stress distribution required for this behavior, we fabricated the LCE actuator through a spring-precursor route (see Methods and Figure S1 for details). A screw mold was used to form the LCE oligomer into a weakly crosslinked soft spring precursor, which was subsequently uniaxially stretched and fixed by further crosslinking. This process converted the spring precursor into a straight fiber while preserving internal asymmetric mechanical programming across the cross-section. After integration of a buoyant bead at the fiber tip, the actuator was able to support buoyancy-coupled non-reciprocal beating.

2.2 | Modes of Non-Reciprocal Self-Oscillation

We next examined the light-driven dynamics of PNCOSs in aqueous environments and their tunability through the position of the incident NIR beam. Depending on the illumination geometry, the system exhibited three representative motion modes: clockwise conical oscillation, counterclockwise conical oscillation, and swing oscillation (Figure 2a–i, Figure S5 and Movies S1–S3). Self-oscillation in aqueous environments occurred only when two conditions were satisfied: the width of the light spot had to be smaller than the light-induced coil diameter, and the light intensity had to exceed the threshold required for partial escape of the fiber from the illuminated region. Otherwise, the deformed fiber remained entirely within the beam and stabilized in a coiled state without oscillation.

The Reynolds numbers for the three periodic motion modes were estimated using the experimentally measured characteristic velocities and length scale, together with the fluid properties of deionized water. The calculated values are approximately 196, 130, and 127 for clockwise conical oscillation, counterclockwise conical oscillation, and planar swinging motion, respectively (Note S3 and Table S1). These values indicate that the PNCOSs operate outside the strict Stokes-flow regime, where both viscous and inertial effects contribute to actuator motion and associated fluid transport.

To further map the oscillatory behavior, we continuously rotated the sample to systematically vary the irradiation position relative to the fiber (Figure 2j). Illumination near the top of the fiber induced clockwise conical oscillation. As the sample was rotated counterclockwise, the motion mode changed accordingly: counterclockwise conical oscillation was observed over $0^\circ < \theta \leq 90^\circ$,

swing oscillation over $90^\circ < \theta < 270^\circ$, and clockwise conical oscillation over $270^\circ \leq \theta \leq 360^\circ$. These results show that both the oscillation trajectory and direction can be regulated in real time through simple adjustment of the light illumination position.

2.3 | Mechanism of Non-Reciprocal Self-Oscillation

To elucidate the mechanism of non-reciprocal self-oscillation in PNCOSs, we first analyzed the photoinduced deformation of the LCE fiber actuator under NIR irradiation. As shown in Figure 3a and Figure S6, localized irradiation first induced bending of the fiber toward the light source, followed by twisting and coiling. Once coiling occurred, the deformed segment moved out of the illuminated region. The buoyant force provided by the tip-mounted foam bead then promoted passive upward recovery of the actuator toward its initial linear state, thereby initiating the next oscillation cycle. This sequence of light-driven deformation, escape from the beam, and buoyancy-assisted recovery generated a closed-loop motion trajectory.

To further resolve the deformation pathway, we used high-speed microscopy to capture the instantaneous response of the fiber under irradiation. Under heavier loading, the fiber underwent sequential twisting, coiling, and shortening. Under lighter loading, it first bent and then transitioned into twisting and coiling (Figure 3b and Figure S7). These observations show that the sequence of deformation depends on the mechanical loading condition.

We next examined how this deformation cycle gave rise to autonomous oscillation in aqueous environments. High-speed imaging showed that NIR irradiation induced local deformation of the actuator and caused part of the fiber to move out of the illuminated region (Figure 3c and Movie S4). After leaving the beam, the deformed segment continued its motion briefly, then cooled and recovered owing to the intrinsic elasticity of the LCE, which returned it toward the irradiated zone. Repeated entry into and exit from the light field therefore established a photothermal-mechanical feedback loop that sustained autonomous oscillation. To further reveal the dynamical origin of the sustained self-oscillation generated by this feedback loop, we established a four-state reduced-order dynamical model, in which the generalized escape displacement, velocity, effective thermal state, and effective mechanical state describe crossing of the illuminated-region boundary, mechanical motion, finite photothermal response, and dynamic stiffness reconstruction, respectively. Linear stability analysis shows that position feedback alone does not necessarily generate periodic motion. When the photothermal-escape coupling is strengthened and crosses the stability boundary, the history dependence and phase lag introduced by the finite thermal and dynamic mechanical-state responses render the partially illuminated equilibrium oscillatory and linearly unstable. This delayed photothermal-mechanical feedback provides positive net mechanical input to the small-amplitude oscillatory mode, acting as effective negative damping in the amplitude dynamics. As the oscillation amplitude increases, the bounded optical switching, velocity-dependent hydrodynamic resistance, and finite state-dependent mechanical response collectively limit further amplitude growth, leading to

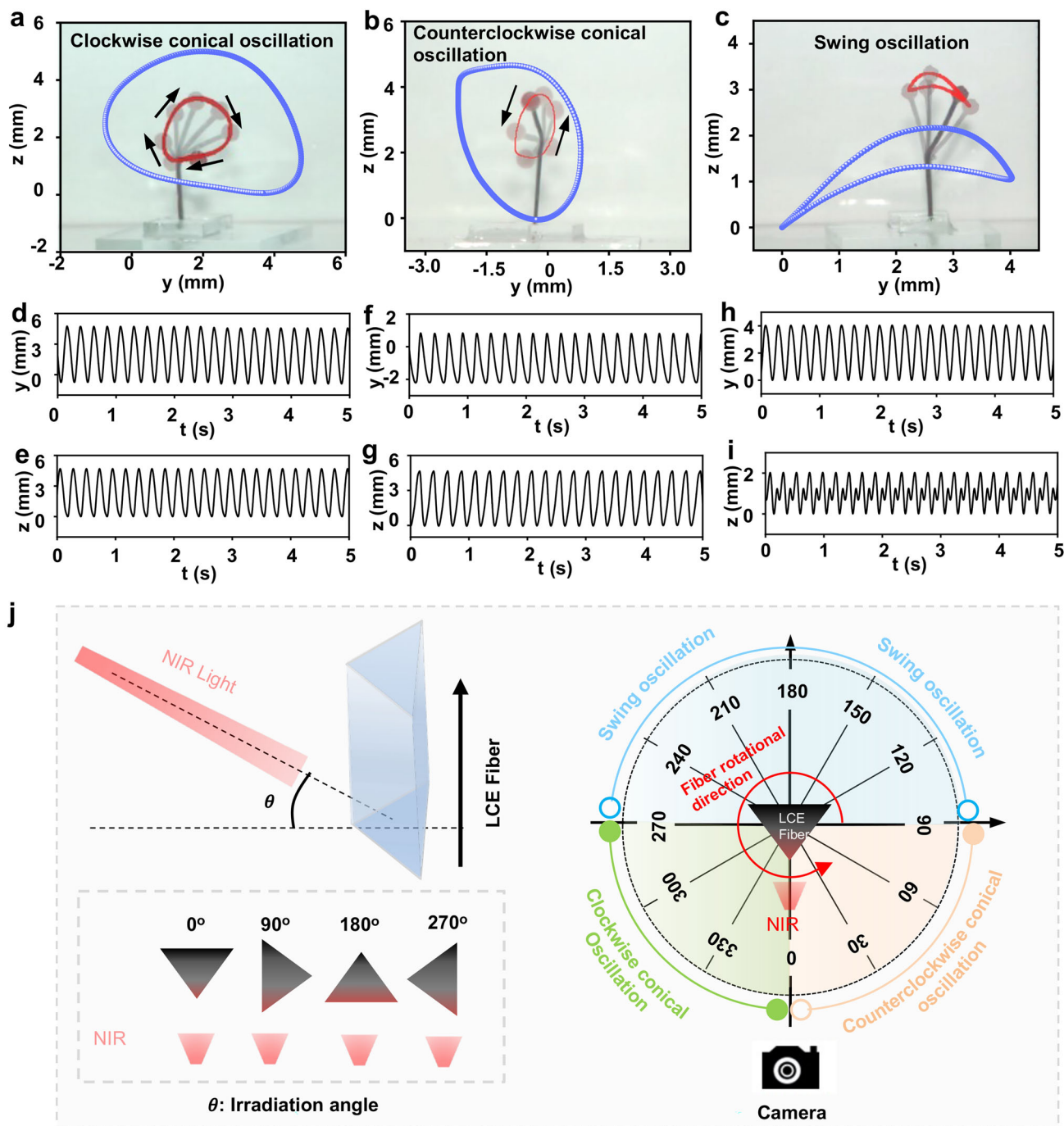


FIGURE 2 | Self-oscillation modes of PNCsOs. a–c) Tip trajectories and corresponding 2D motion plots for three representative oscillation modes: clockwise conical oscillation(a), counterclockwise conical oscillation (b), and swing oscillation (c). d–i) Time-dependent tip displacements along the y and z directions for the three oscillation modes. The LCE fiber length was 20 mm, and the NIR light intensity was 0.85 W. j) Dependence of self-oscillation mode on the irradiation angle. The left panel illustrates the relationship between the incident NIR light direction and the fiber angle, with the light direction aligned with the camera viewing direction. The right panel shows the self-oscillation mode induced at different irradiation angles.

nonlinear saturation and eventual numerical convergence to an attracting stable limit cycle (Note S1 and Figures S8 and S9). Thus, the sustained oscillation of the PNCsO is not guaranteed merely by the existence of a feedback loop, but arises from the oscillatory instability of the partially illuminated equilibrium followed by nonlinear amplitude saturation. To confirm that this behavior originated from motion in and out of the light field rather than self-shadowing by the opaque foam bead, we

replaced the bead with a transparent polystyrene sheet. The system showed the same self-oscillatory behavior, while allowing direct visualization of the fiber moving into and out of the illuminated region (Figure S10 and Movie S5). Non-reciprocal beating further required distinct actuation and recovery paths. Natural cilia achieve such asymmetry through phase-dependent stiffness modulation, in which the effective stroke is stiffer than the recovery stroke [34]. In our system, uniaxially aligned LCE

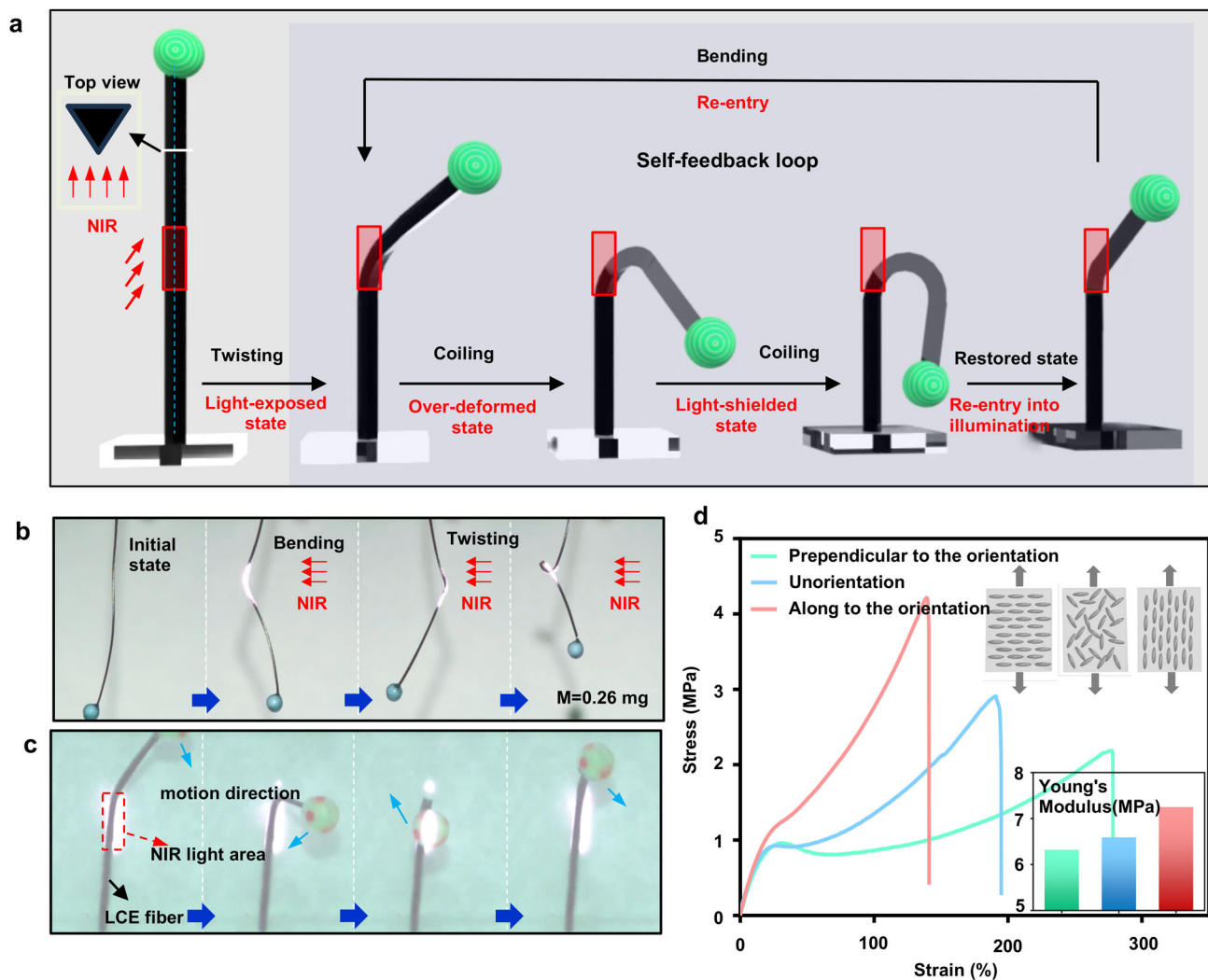


FIGURE 3 | Mechanism of non-reciprocal self-oscillation. a) Schematic illustration of the deformation pathway and morphology evolution of the PNCISO under NIR irradiation. b) High-speed camera images showing the transformation of an LCE fiber from a straight to a helical configuration under NIR irradiation. c) High-speed camera snapshots showing the irradiated PNCISO moving into and out of the illuminated region during clockwise conical oscillation. The cyclic motion is visualized through the position of the shadow cast by the irradiated actuator within the light spot. The LCE fiber length was 20 mm, and the NIR light intensity was 0.85 W. d) Stress–strain curves of LCE films with liquid crystal mesogens aligned parallel to, perpendicular to, or without alignment relative to the stretching direction. The inset at the upper right illustrates the mesogen arrangements in the different film configurations, and the inset at the lower right summarizes the corresponding Young's moduli.

fibers initially possessed a highly ordered mesogen arrangement, and NIR irradiation disrupted this order, thereby reducing the effective stiffness of the material. To verify this mechanism, we measured the mechanical properties of monodomain LCE films with mesogens aligned parallel to, perpendicular to, or without alignment relative to the stretching direction (Figure 3d). Films with parallel alignment exhibited the highest modulus and tensile strength, followed by unaligned films and then perpendicularly aligned films. These results confirm that the degree of mesogen order strongly influences stiffness, supporting the idea that light-induced disordering softens the actuator during irradiation. In combination with the closed-loop deformation–recovery cycle and buoyancy at the fiber tip, this stiffness modulation generated asymmetric actuation and recovery paths, leading to cilia-like non-reciprocal self-oscillation.

2.4 | Tunable Oscillatory Performance and Operational Stability

Having established the mechanism of non-reciprocal self-oscillation, we next examined how geometric and external parameters tune the oscillatory performance of PNCISOs. The oscillation frequency and swept area depended primarily on fiber length, buoyancy, and light intensity. As shown in Figure 4a,b and Figure S11, longer fibers exhibited lower oscillation frequencies and larger swept areas. This trend is qualitatively consistent with previous observations that, once periodic ciliary beating is established, the beat frequency tends to decrease slightly with increasing cilium length [35]. In this system, the length dependence likely reflects a slower mechanical response under localized photothermal actuation. Because the illuminated region

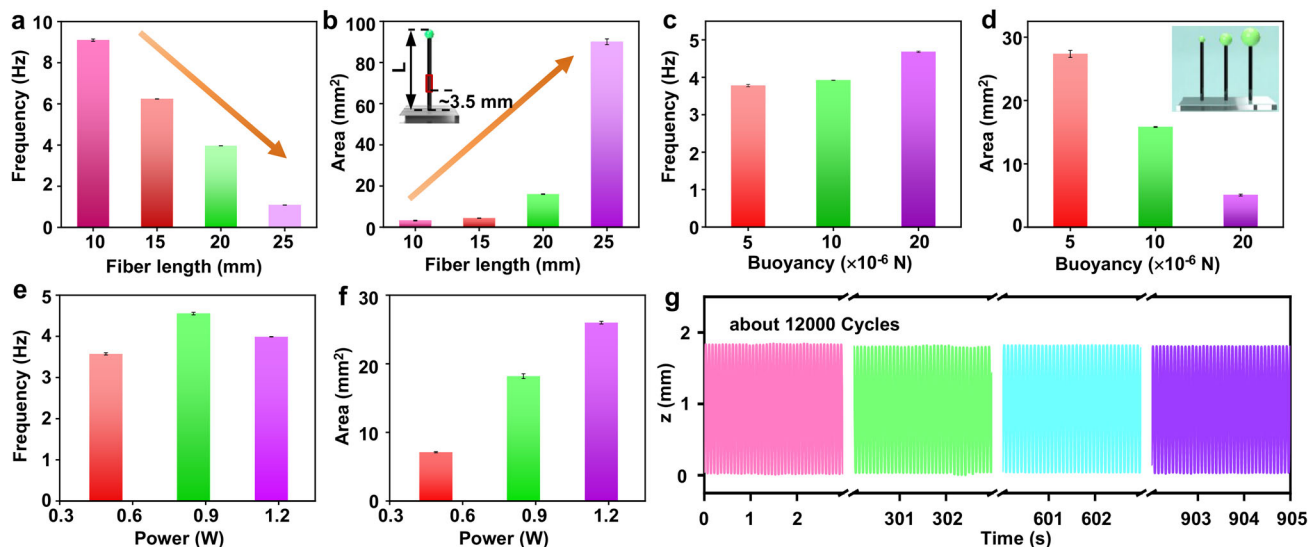


FIGURE 4 | Tunable oscillatory performance and operational stability of PNCsOs. a, b) Oscillation frequency (a) and swept area (b) as functions of fiber length. c, d) Oscillation frequency (c) and swept area (d) as functions of buoyancy. e, f) Oscillation frequency (e) and swept area (f) as functions of NIR light power. Data are presented as mean $\pm$ SEM ($n = 3$), calculated from representative oscillation cycles selected from the beginning, middle, and end of each recording. g) Tip displacement along the z direction during prolonged oscillation under 0.85 W NIR irradiation.

is similar for short and long fibers, the reduced oscillation frequency of longer fibers is more plausibly associated with their larger effective lever arm and slower deformation–recovery dynamics. The oscillation frequency reached 23.1 Hz at a fiber length of 4.2 mm and 31.7 Hz at 2.2 mm (Figure S12). By contrast, buoyancy and light intensity had relatively minor effects on oscillation frequency but strongly affected the swept area, which decreased with increasing buoyancy and increased with increasing light intensity (Figure 4c–f). To further clarify the controlling scale of the oscillation period, we performed a relative response analysis of the frequency data in Figure 4 in logarithmic coordinates. The experimental relative frequency–response coefficients corresponding to fiber length, optical power, and buoyancy are, respectively,

$$S_L^{\text{exp}} = -2.156, S_P^{\text{exp}} = 0.120, S_B^{\text{exp}} = 0.145$$

The absolute value of the length–response coefficient is approximately 18 and 15 times those of the optical–power and buoyancy responses, respectively, indicating that the oscillation frequency responds much more strongly to fiber length as a combined mechanical/geometric control variable than to external optical input and buoyancy. Consistently, local sensitivity analysis of the reduced-order dynamical model shows that the effective inertia and restoring stiffness are the internal parameters to which the frequency is most sensitive, while thermal relaxation provides a secondary but non-negligible correction to the period and optical input and buoyancy produce weaker frequency variations (Note S2 and Figure S13). These results indicate that the oscillation frequency of the PNCsO is governed primarily by the overall inertia–restoring mechanical timescale, while finite photothermal relaxation further modifies the limit-cycle period by altering the feedback phase relationship; by contrast, optical power and buoyancy mainly regulate the motion amplitude and trajectory.

PNCsOs also exhibited stable long-term operation. We further evaluated the operational stability of PNCsOs under prolonged cycling. After 12,000 consecutive oscillation cycles, no significant change was observed in either tip displacement or oscillation frequency (Figure 4g), indicating stable long-term self-oscillation.

2.5 | Adaptive and Synchronized Collective Motion of PNCsOs

We then asked whether the cilia-like actuator could adapt its oscillatory behavior to multiple optical inputs [36]. As shown in Figure 5a, irradiation from the left induced counterclockwise conical oscillation. Introducing a second light source from the front altered both the oscillation trajectory and frequency, decreasing the frequency from 7.07 to 5.82 Hz while increasing the swept area from 0.96 to 4.01 mm². After removal of the second light source, the PNCsO gradually returned to its original self-oscillation mode (Figure 5a,b). Similar responses were observed under other illumination configurations (Figure S14 and Movie S6): a second light source from the left maintained counterclockwise conical oscillation, whereas a second light source from the right induced clockwise conical oscillation. These results show that the PNCsO can integrate multiple optical inputs and continuously reconfigure its oscillatory motion in response to changes in the external light field.

We next investigated whether two adjacent PNCsOs could exhibit synchronized motion through dynamic coupling. Previous studies have shown that synthetic oscillators can synchronize through substrate-mediated mechanical interactions in air [37]. In the present submerged system, coupling could be mediated by the shared substrate, the surrounding liquid, or both. Two PNCsOs were mounted on a common 100 μm -thick flexible PDMS film. When stimulated separately, the red-ball PNCsO exhibited clockwise conical oscillation at 5.99 Hz, whereas the green-ball PNCsO

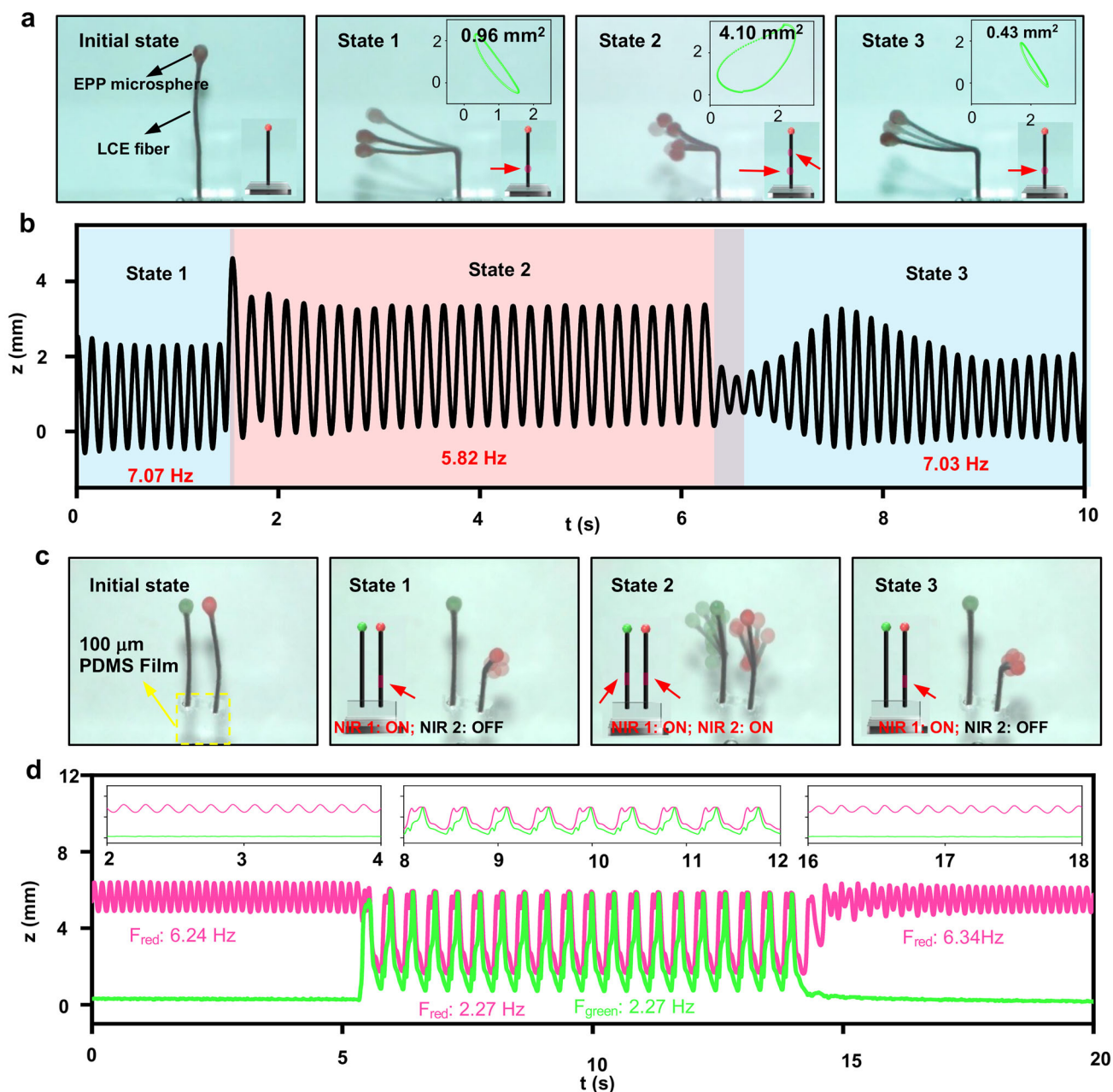


FIGURE 5 | Adaptive, synchronized, and functional motion of PNCsOs. a) Sequential optical images showing adaptive reconfiguration of PNCsO motion under multiple NIR inputs. In the first state, irradiation from the left induced oscillation at approximately 7.07 Hz. In the second state, an additional NIR beam from the front changed the motion to counterclockwise conical oscillation at 5.82 Hz. In the third state, after removal of the second light source, the original oscillation mode was restored. b) Tip displacement of the PNCsO along the z direction under sequential NIR illumination. c) Optical images of two neighboring PNCsOs connected by a 100 μm PDMS film. Simultaneous irradiation of the two PNCsOs with separate NIR beams induced synchronized coupled oscillation. d) Tip displacement of the coupled PNCsOs along the z direction under NIR irradiation.

displayed counterclockwise conical oscillation at 2.27 Hz (Figure S15). In the sequential-activation experiment, the red-ball PNCsO initially oscillated alone at 6.24 Hz and slowed to approximately 2.2 Hz after activation of the green-ball PNCsO. During the coupled stage, the two PNCsOs exhibited a shared dominant frequency, and their phase difference stabilized at $12.1^\circ \pm 2.1^\circ$ (Figure S16). The coincident fundamental frequencies, together with the stable phase difference, supported stable 1:1 frequency and phase locking, whereas the higher-order spectral peaks were attributed to harmonics of the nonsinusoidal displacement

waveforms. After deactivation of the green-ball PNCsO, the red-ball PNCsO recovered a high-frequency motion state comparable to its initial state (Figure 5c,d and Figure S16 and Movie S7), demonstrating reversible dynamic coupling between neighboring PNCsOs.

To examine whether synchronization depended exclusively on deformation of the flexible PDMS substrate, we performed a control experiment using a rigid glass substrate. The initially active PNCsO oscillated at 16.5 Hz, shifted to a common

frequency of 15.2 Hz after activation of its neighbor, and recovered to 16.6 Hz after the neighboring PNCSO stopped. During the coupled interval, the two PNCSOs maintained a stable phase lag of $59.2^\circ \pm 2.9^\circ$ and exhibited coincident dominant frequencies (Figure S17). The persistence of reversible frequency and phase locking on glass shows that coupling does not rely solely on deformation of the flexible PDMS substrate and supports a contribution from fluid-mediated hydrodynamic interactions to the coupled motion. Mechanical transmission through the glass, adhesive layer, or mounting structure may additionally contribute.

Coupling was also observed between PNCSOs of different lengths. During simultaneous activation, the 10 and 15 mm PNCSOs entered a common low-frequency oscillatory state at 1.90 and 1.92 Hz, respectively, with a stable phase lag of $79.1^\circ \pm 1.0^\circ$ (Figure S18). This coupled state was accompanied by substantial changes in frequency and amplitude relative to the isolated motions, showing that PNCSOs with different individual dynamics can form a shared oscillatory state. Because the same PDMS film was present before, during, and after activation of the neighboring PNCSO, the reversible frequency changes cannot be attributed solely to a static change in effective actuator length or boundary condition. Collectively, these results indicate that both substrate-mediated mechanical interactions and fluid-mediated hydrodynamic interactions can affect the coupling between neighboring PNCSOs.

Under the conditions examined here, the coupled PNCSOs maintained periodic motion with stable phase lags and showed no continuous phase drift or additional incommensurate frequency peaks during the analyzed intervals. The higher-order spectral peaks were harmonics associated with the nonsinusoidal displacement waveforms rather than evidence of quasiperiodic dynamics. Nevertheless, more complex dynamical regimes, including quasiperiodic or irregular motion, may emerge under larger intrinsic-frequency detuning, different coupling strengths, or conditions near the self-oscillation stability boundary. A systematic detuning–coupling phase diagram is beyond the scope of the present study and represents an important direction for future investigation.

2.6 | Motion Mode Dependent Particle Manipulation by PNCSOs

The ability of PNCSOs to generate non-reciprocal self-oscillation in fluidic environments also enabled active manipulation of nearby objects through fluid–structure interactions. To examine how the actuator motion mode influences particle transport, polystyrene microparticles were dispersed in 10 cSt silicone oil. The silicone oil was selected to increase viscous resistance and decrease the particle sedimentation rate, thereby facilitating continuous trajectory tracking during the observation period. Three representative oscillation modes were investigated: clockwise conical oscillation, counterclockwise conical oscillation, and planar swinging motion (Figure 6a–f and Movie S8).

Clockwise and counterclockwise conical oscillations generated circulating particle motion in opposite angular directions. To quantify this behavior, four to five representative particles were

tracked under each condition, and their cumulative angular displacements relative to the projected actuator base were calculated (Figure S19). Most particles exhibited angular displacements consistent with the rotational direction of the actuator, although their detailed trajectories and displacement magnitudes varied depending on their initial positions, radial distances from the actuator, and transient interactions with the local flow. In contrast, planar swinging primarily generated localized back-and-forth particle motion, resulting in smaller and more variable net angular displacements. These results demonstrate that the direction and qualitative pattern of particle transport can be selected by changing the oscillation mode of the PNCSO.

As a complementary demonstration involving a larger object, a polystyrene particle was introduced into a water–glycerol mixture (1:1 by volume) to reduce sedimentation and facilitate motion tracking. Upon illumination, the PNCSO exhibited counterclockwise non-reciprocal self-oscillation and drove the nearby particle to undergo repeated counterclockwise orbital motion (Figure 6g and Movie S9). Particle motion induced by a PNCSO was also observed in 20 cSt silicone oil, qualitatively demonstrating that the manipulation capability persists at a higher liquid viscosity (Movie S10).

These observations demonstrate that the self-sustained motion of PNCSOs generates localized fluid flows that produce motion-mode-dependent particle transport without externally programmed mechanical actuation. These results represent a proof-of-concept demonstration of directional particle manipulation rather than deterministic routing or closed-loop positioning of individual particles.

The present PNCSOs have characteristic dimensions in the millimeter-to-centimeter range, allowing direct visualization of self-sustained nonreciprocal oscillation, actuator–actuator coupling, and object manipulation in water. These dimensions are larger than those of many artificial cilia developed for microfluidic applications, including magnetic and light-responsive cilia based on liquid-crystal polymers or networks at micrometer-to-submillimeter scales [19, 38–40]. These studies demonstrate that cilia-like structures can be fabricated at substantially smaller dimensions. However, because PNCSOs function as coupled self-oscillating systems rather than externally driven actuators, their oscillatory behavior cannot be maintained through geometric scaling alone. Miniaturization would require rebalancing the buoyant force, LCE fiber stiffness and programmed self-winding stress, viscous resistance, photothermal response, and position-dependent illumination. In particular, buoyancy decreases with the volume of the attached bead, while optical addressing and shadowing become more sensitive as the displacement amplitude decreases. Fabrication of self-winding LCE microfibers with controlled internal stress and integration of a sufficiently buoyant microelement would also require further development. Downscaling toward microfluidic dimensions would therefore require coordinated optimization of the fiber geometry, material modulus, buoyant element, photothermal filler content, illumination profile, and fluidic conditions. At smaller scales, arrays of PNCSOs may be more suitable for pumping and mixing than individual actuators for manipulating relatively large objects.

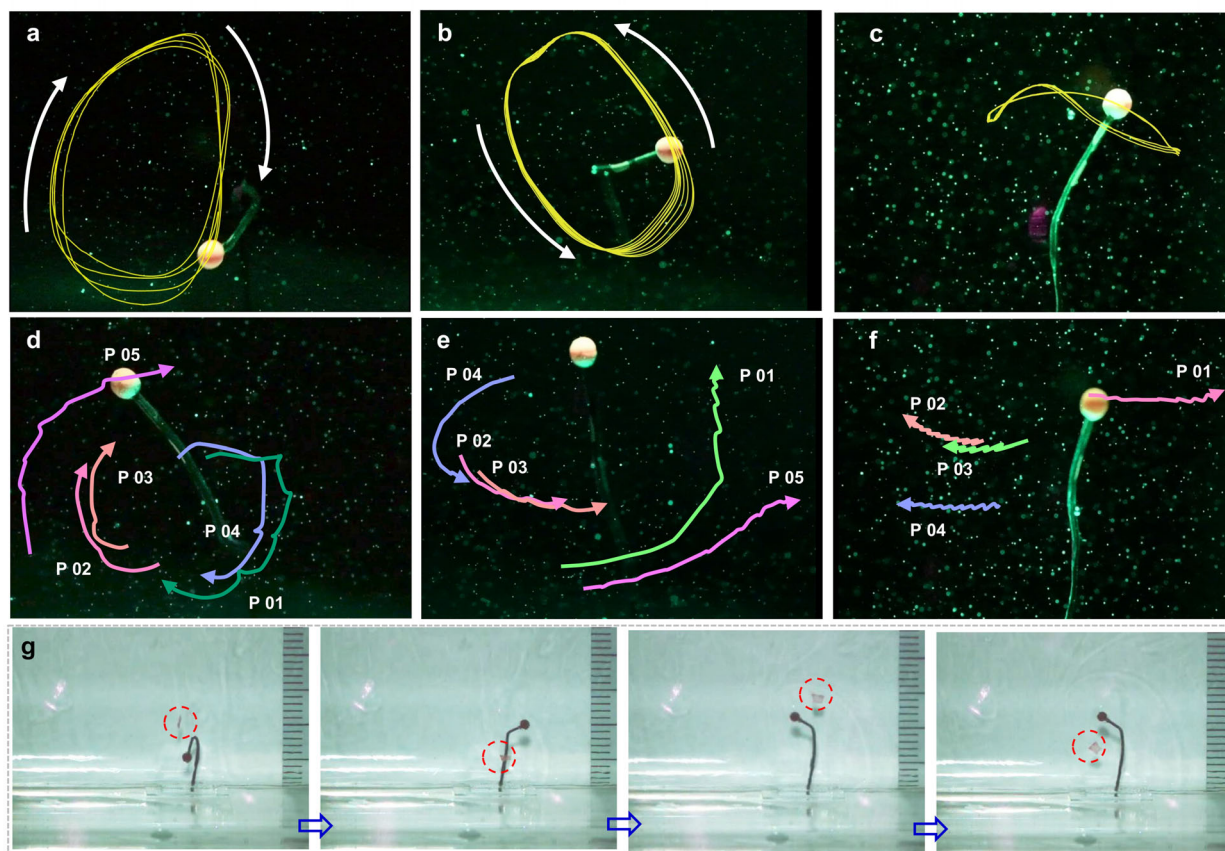


FIGURE 6 | Motion-mode-dependent particle manipulation by the PNCsO. a–c) Representative trajectories of actuator tip during clockwise conical oscillation, counterclockwise conical oscillation, and planar swinging motion, respectively. The white arrows indicate the motion direction of actuator. d–f) Trajectories of four or five representative tracer particles, labeled P01–P05 as applicable, under the corresponding actuator motions. The tracer particles were dispersed in 10 cSt silicone oil. The colors of the particle trajectories correspond to those used in the quantitative plots in Figure S19, and the arrowheads indicate the motion direction of particle. Clockwise and counterclockwise conical oscillations generate particle circulation in opposite angular directions, whereas planar swinging motion primarily induces localized back-and-forth displacement of particles. g) Time-lapse images showing the repeated displacement of a larger polystyrene particle by the oscillating actuator. The red dashed circles indicate the position of the transported particle, and the blue arrows indicate the temporal sequence.

3 | Conclusions

In summary, we report phototunable nonreciprocal cilia-like self-oscillators (PNCsOs) that integrate self-sustained oscillation, non-reciprocal motion, multimode tunability, and adaptive behavior in aqueous environments. By integrating a self-winding LCE fiber actuator, tip-induced buoyancy, and localized NIR irradiation, PNCsOs reproduce several dynamic features of natural cilia, including sustained oscillation, asymmetric beating trajectories, and adaptive responses to changing external inputs. The oscillatory behavior can be tuned in real time through light position, light intensity, fiber length, and buoyancy, and adjacent PNCsOs can further display synchronized motion through coupling. In addition, the non-reciprocal self-oscillation of PNCsO enables active manipulation of small objects in viscous aqueous environments. It is well known that synchronization and collective motion are a captivating expression of self-organization that nature has been leveraged to sustain life, and replicating such properties in artificial active materials has been a great challenge for all length scales [37]. Intriguingly, we have discovered that two adjacent PNCsOs

enable communication together and synchronization of their self-oscillating motions, resembling the natural expression of self-organization. These results establish a versatile strategy for constructing adaptive and functional artificial ciliary systems. Future work will focus on improving design efficiency, extending coupled interactions among multiple PNCsOs, and integrating these systems into more complex soft robotic and microfluidic platforms.

4 | Experimental Section

4.1 | Materials

1,4-Bis-[4-(6-acryloyloxyhexyloxy)benzoyloxy]-2-methylbenzene (RM82) and 4-(6-(acryloyloxy)hexyloxy)phenyl-4-(6-(acryloyloxy)hexyloxy)benzoate (C6BAPE) was purchased from Henan Daken Chemical Co., Ltd. DODT (3,6-dioxo-1,8-octane dithiol), pentaerythritol tetrakis (3-mercaptopropionate) (PETMP), dipropylamine (DPA), and graphene were obtained from Sigma-Aldrich.

4.2 | Characterizations

NIR light (808 nm) was generated by a laser pointer (HW808AD1200-22FGD, Shenzhen Infrared Laser Technology Co., Ltd) or a laser source (PSU-H-LED, MDL-H-808-5 W). The laser intensity was monitored by a laser power meter (TP100, Changchun New Industries Optoelectronics Technology Co., Ltd). Supplementary videos were recorded by a super-resolution digital microscope (Keyence, VHX-1000C) or a digital camera (Canon, EOS 80D(W)). Slow motions of self-oscillations were recorded by a high-speed camera (Keyence VW-9000). Self-oscillating motions were tracked by kinematic analysis software (Kinovea). DSC thermograms were obtained by TA Discovery DSC 250 (rate of heating: 5°C min⁻¹, N₂ flow 50 mL min⁻¹). Mechanical properties of the LCE fibers were investigated by Instron 5943 universal testing system with the tensile testing rate of 10 mm min⁻¹.

4.3 | Preparation of the LCE Soft Spring Precursor for PNCOSOs

A screw-shaped mold was used to prepare reactive fibrous precursors for cilia-like self-oscillator. RM82, C6BAPE, DODT, and PETMP were mixed at a molar ratio of 3:1:2.4:0.8, with equimolar thiol and acrylate groups. Graphene was added at 2 wt%, and the mixture was dissolved in chloroform. After sonication for 4 h to disperse the graphene uniformly, a catalytic amount of dipropylamine (DPA) was added to initiate the thiol–Michael addition reaction. The resulting mixture was then injected into the screw-shaped mold and allowed to react at room temperature for 2 h, yielding a weakly crosslinked LCE soft spring precursor.

4.4 | Preparation of the Self-Winding LCE Actuator for PNCOSOs

The weakly crosslinked LCE soft spring precursor was removed from the mold, mechanically drawn, and untwisted into a straight fiber. The straightened fiber was subsequently stretched to 50% strain and held at this strain for 24 h to complete the crosslinking reaction and fix uniaxial molecular alignment. After curing, a uniaxially aligned LCE fiber actuator was obtained. Owing to the programmed internal stress generated during this processing route, the resulting fiber served as the self-winding active ciliary unit of the PNCOSOs.

Author Contributions

J.L. conceived the research. Z.H. and J.L. designed the experiments. Z.H., T.Z., J.W., and X.S. carried out the experiments. J.L. and Z.H. analyzed the experimental data. Z.H. and K.L. analyzed the oscillation mechanism. Z.H. and J.L. wrote the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supporting Information.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: adfm78414-sup-0001-SuppMat.pdf.

Supporting File 2: adfm78414-sup-0002-MovieS1.mp4.

Supporting File 3: adfm78414-sup-0003-MovieS2.mp4.

Supporting File 4: adfm78414-sup-0004-MovieS3.mp4.

Supporting File 5: adfm78414-sup-0005-MovieS4.mp4.

Supporting File 6: adfm78414-sup-0006-MovieS5.mp4.

Supporting File 7: adfm78414-sup-0007-MovieS6.mp4.

Supporting File 8: adfm78414-sup-0008-MovieS7.mp4.

Supporting File 9: adfm78414-sup-0009-MovieS8.mp4.

Supporting File 10: adfm78414-sup-0010-MovieS9.mp4.

Supporting File 11: adfm78414-sup-0011-MovieS10.mp4.